



Hardened Segmented Network Evidence Portfolio

Project 3 - pfSense Segmentation, Admin Isolation, and Monitoring

Target: segmented pfSense firewall lab with WAN, WEB, DB, and ADMIN zones

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Assessment Type: Network Architecture, Firewall Policy, Traffic Validation, Admin Access Hardening, Logging, Monitoring

Document Classification: Public Portfolio Sample / Capability Demonstration

Prepared For: Portfolio Review / Technical Assessment

Date: September 13, 2026

Evidence Screenshots: 42

Assessment Environment Note

This portfolio evidence was completed in a controlled local lab workstation environment. pfSense, Kali clients, VirtualBox networking, screenshots, logs, and validation steps were used to keep the evidence reproducible, auditable, and easy to verify.

Portfolio Boundary

This report demonstrates hardened network segmentation and controlled local firewall administration. It is not a live production engagement or scan of any real company environment. Live validation would require written authorization, defined scope, staging infrastructure, monitoring, and stop conditions.

Project 3 Main Topic

A segmented pfSense firewall lab proving that WEB, DB, and ADMIN zones can be separated, lateral movement can be blocked, firewall management can be isolated to ADMIN, and security events can be logged and monitored.

Evidence Index

The sections below preserve the screenshot folder order from the Project 3 evidence directory.

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Firewall Installation and Baseline

2 evidence screenshot(s)

Test 02-Network-Architecture-and-Zones

Network Architecture and Zones

5 evidence screenshot(s)

Test 03-Interface-Assignments-and-DHCP

Interface Assignments and DHCP

11 evidence screenshot(s)

Test 04-Firewall-Rules-and-Segmentation-Policy

Firewall Rules and Segmentation Policy

7 evidence screenshot(s)

Test 05-Allowed-Traffic-Validation

Allowed Traffic Validation

4 evidence screenshot(s)

Test 06-Blocked-Lateral-Movement-Validation

Blocked Lateral Movement Validation

3 evidence screenshot(s)

Test 07-Admin-Access-Hardening

Admin Access Hardening

5 evidence screenshot(s)

Test 08-Logging-and-Monitoring-Evidence

Logging and Monitoring Evidence

4 evidence screenshot(s)

Test 09-Final-Risk-Reduction-Summary

Final Risk Reduction Summary

1 evidence screenshot(s)

Test 01-Firewall-Installation-and-Baseline

Firewall Installation and Baseline

Control Objective

Validated pfSense installation, system identity, version, and healthy WAN/LAN baseline.

Evidence screenshots in this section: 2

- 1A-pfsense-System Identity-Version
- 1B-pfsense-Health WAN-LAN Baseline

Firewall Installation and Baseline

1A pfsense System Identity Version

 pfSense
COMMUNITY EDITION



Dashboard  

System Information   

Name	pfSense.p3lab.local
User	admin@10.10.10.100 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 266620c497dea6c15836
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.9.0-RELEASE (amd64) built on Fri Aug 28 19:53:00 CEST 2026 FreeBSD 16.0-CURRENT Unable to check for updates 
CPU Type	Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz 2 CPUs : 1 package(s) x 2 cache groups x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Enabled
MDS Mitigation	Inactive
Uptime	05 Hours 13 Minutes 10 Seconds
Current date/time	Fri Sep 11 22:40:18 CEST 2026

Firewall Installation and Baseline

1B pfsense Health WAN LAN Baseline

OffSecKali LinuxKali ToolsKali DocsKali ForumsKali NetHunterExploit-DB>>

• 1.1.1.1

• 8.8.8.8

Last config change

Fri Sep 11 21:38:31 CEST 2026

State table size

0% (28/403000)

Show states

MBUF Usage

0% (4572/1000000)

Load average

0.34, 0.54, 0.58

CPU usage

13%

Memory usage

9% of 4036 MiB

SWAP usage

0% of 1024 MiB

Disks

⚙️⌵⌵

Mount	Used	Size	Usage
> /	957M	24G	4% of 24G (zfs)

Netgate Services And Support

⌵⌵

Retrieving support information ⚙️

↺

Interfaces

⚙️⌵⌵

🏠 WAN	↑	1000baseT <full-duplex>	10.0.2.15 fd17:625c:f037:2:c44a:803c:1451:efc5
🏠 LAN	↑	1000baseT <full-duplex>	10.10.10.1

Test 02-Network-Architecture-and-Zones

Network Architecture and Zones

Control Objective

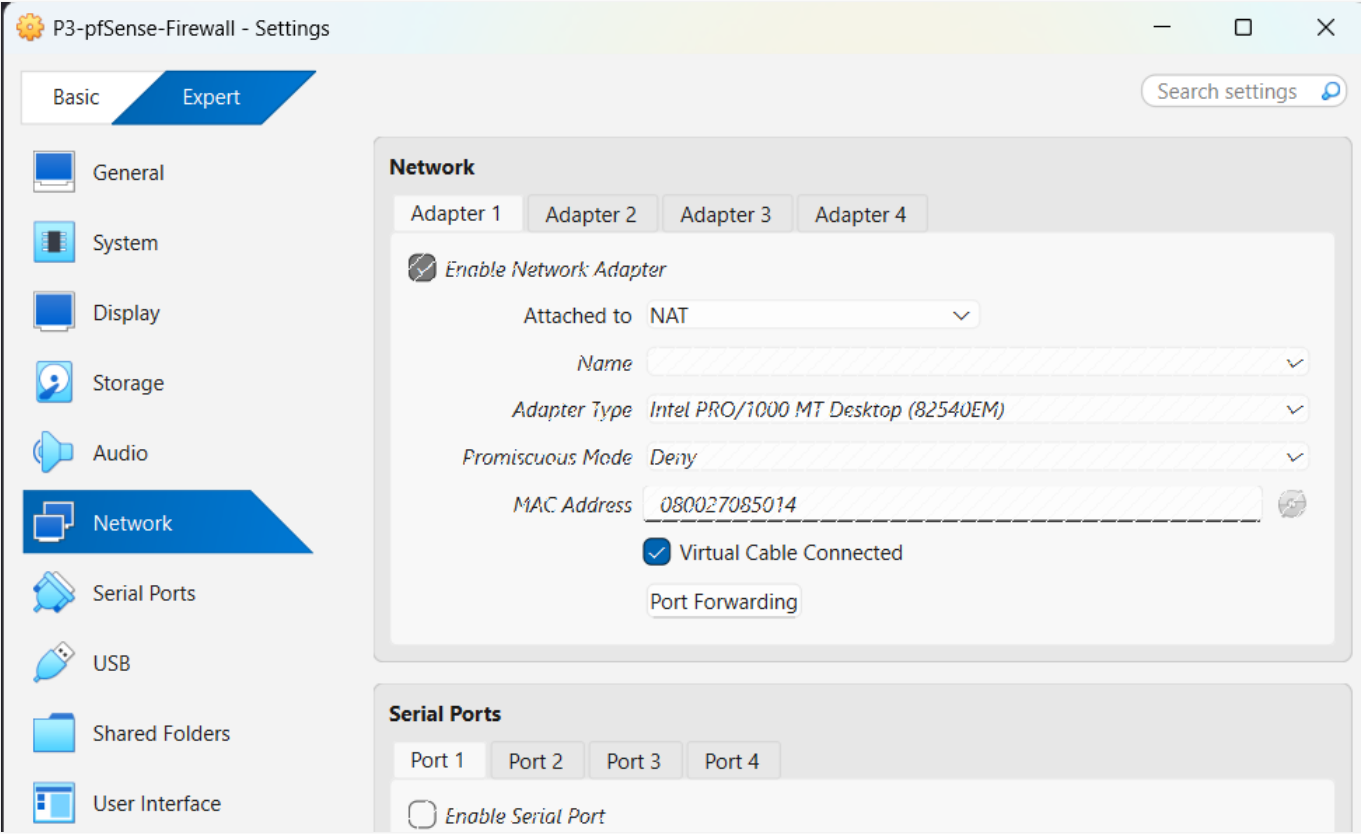
Documented the segmented VirtualBox architecture for WAN, WEB, DB, and ADMIN zones.

Evidence screenshots in this section: 5

- 2A-pfsense-adapter1-wan-nat
- 2B-pfsense-adapter2-web-zone
- 2C-pfsense-adapter3-db-zone
- 2D-pfsense-adapter4-admin-zone
- 2E-project3-zone-ip-plan

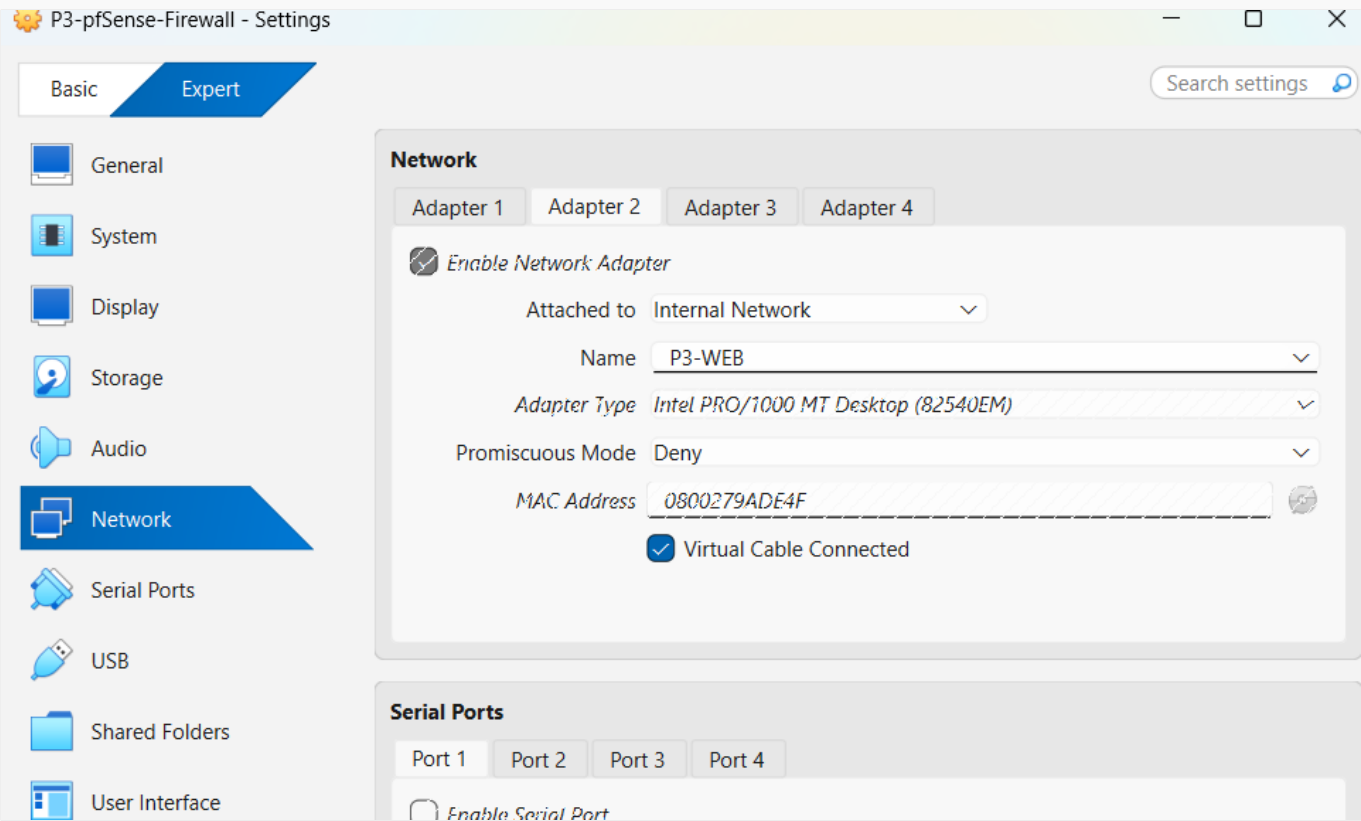
Network Architecture and Zones

2A pfsense adapter1 wan nat



Network Architecture and Zones

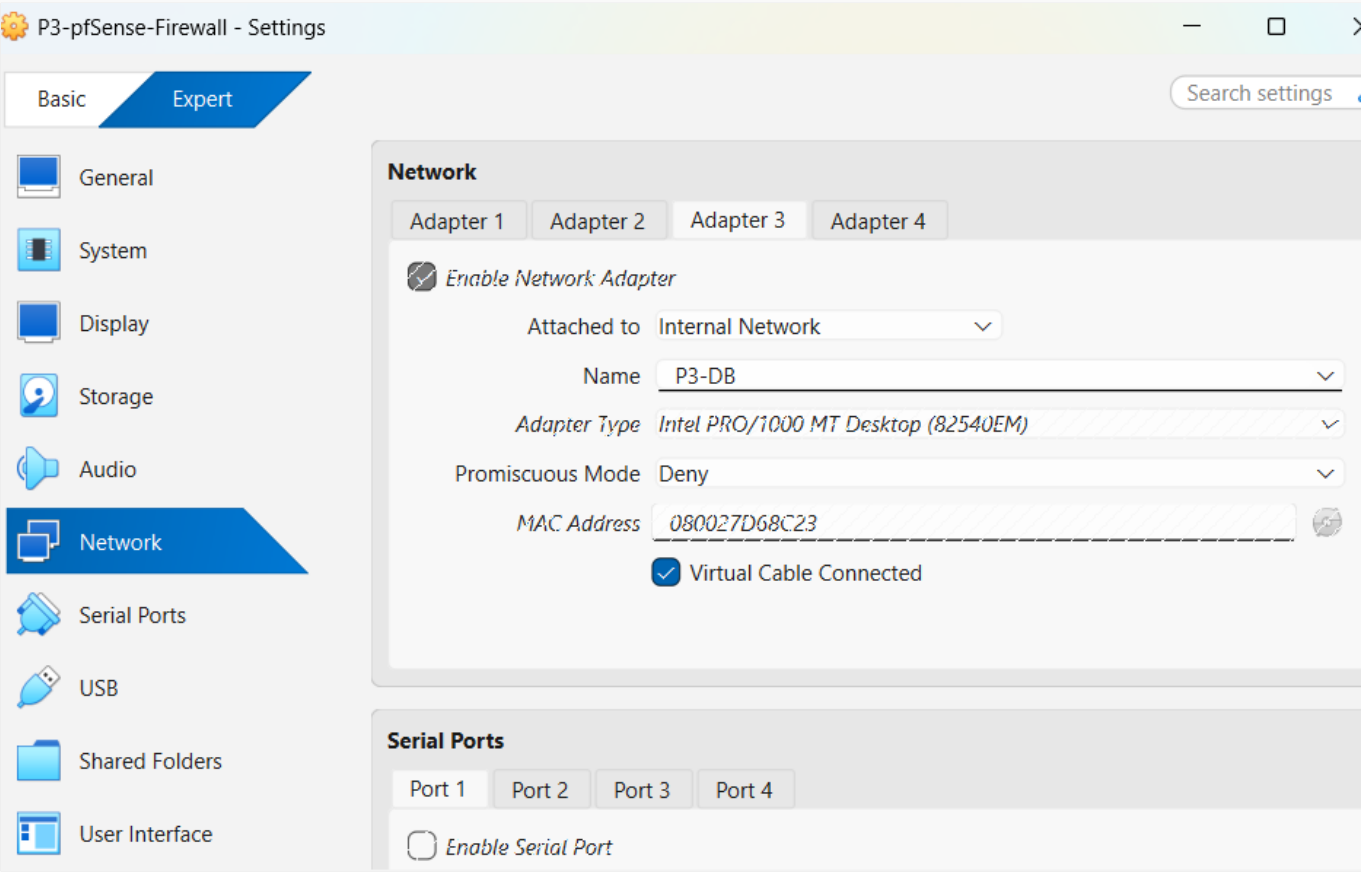
2B pfSense adapter2 web zone



2B-pfsense-adapter2-web-zone.png

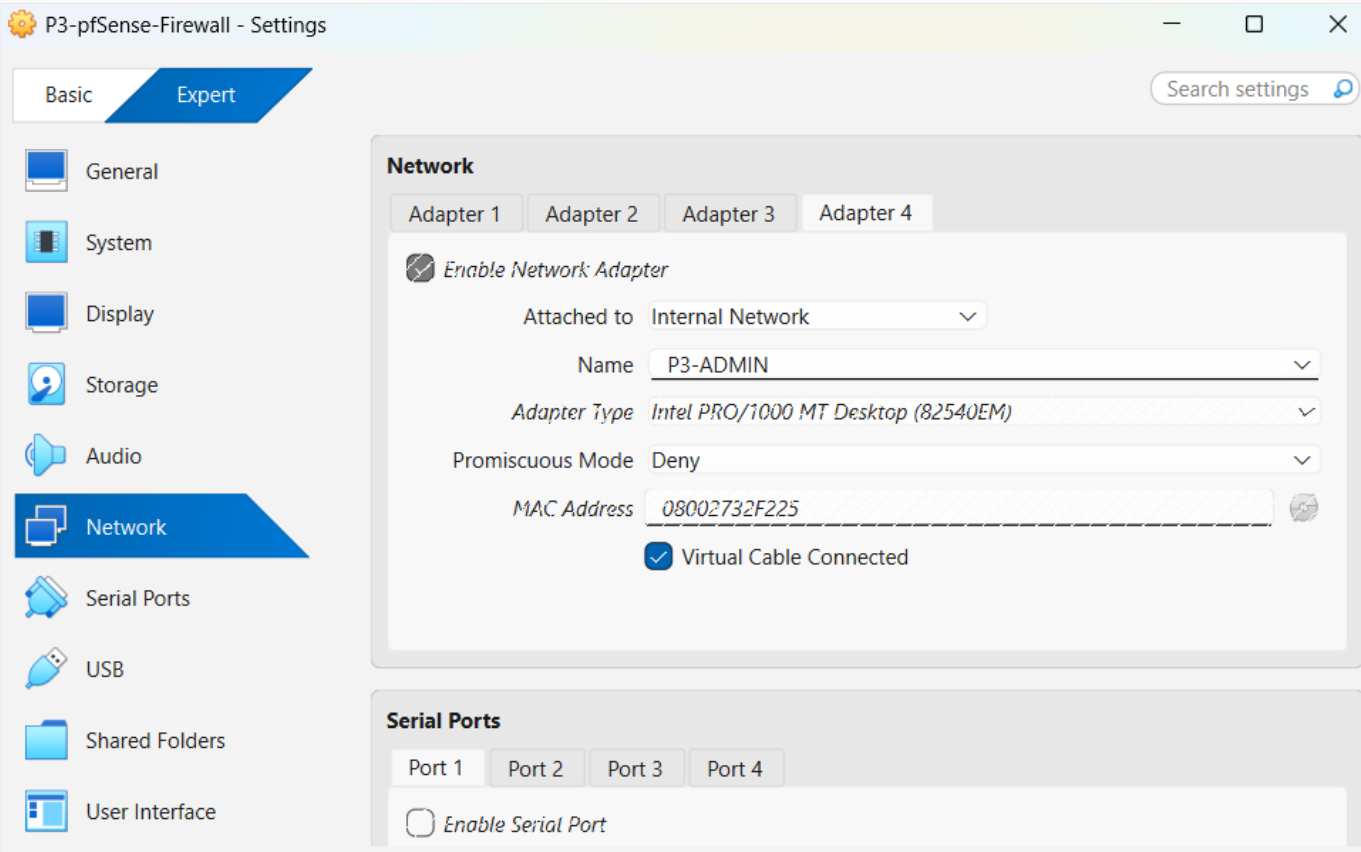
Network Architecture and Zones

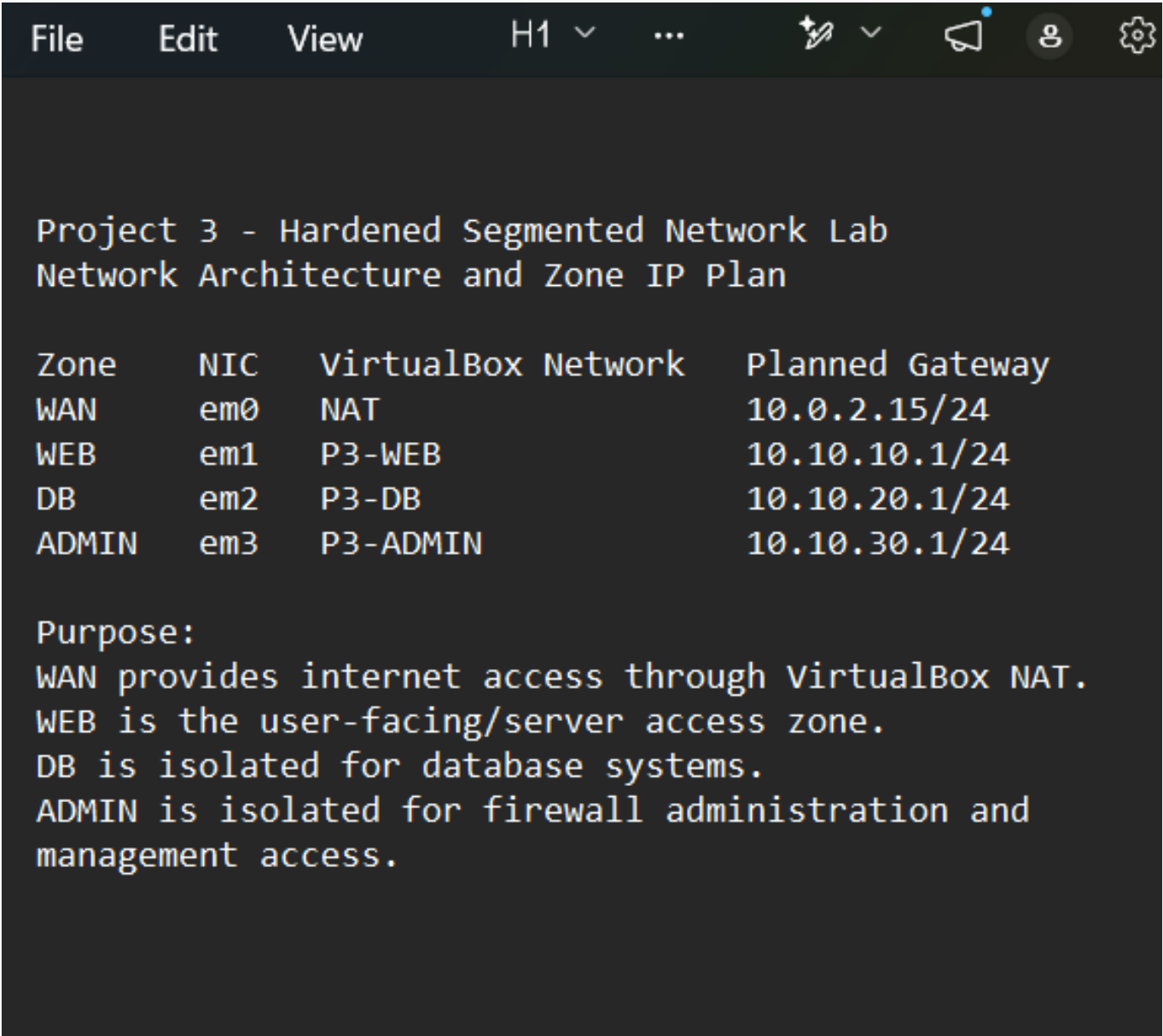
2C pfsense adapter3 db zone



Network Architecture and Zones

2D pfSense adapter4 admin zone





Test 03-Interface-Assignments-and-DHCP

Interface Assignments and DHCP

Control Objective

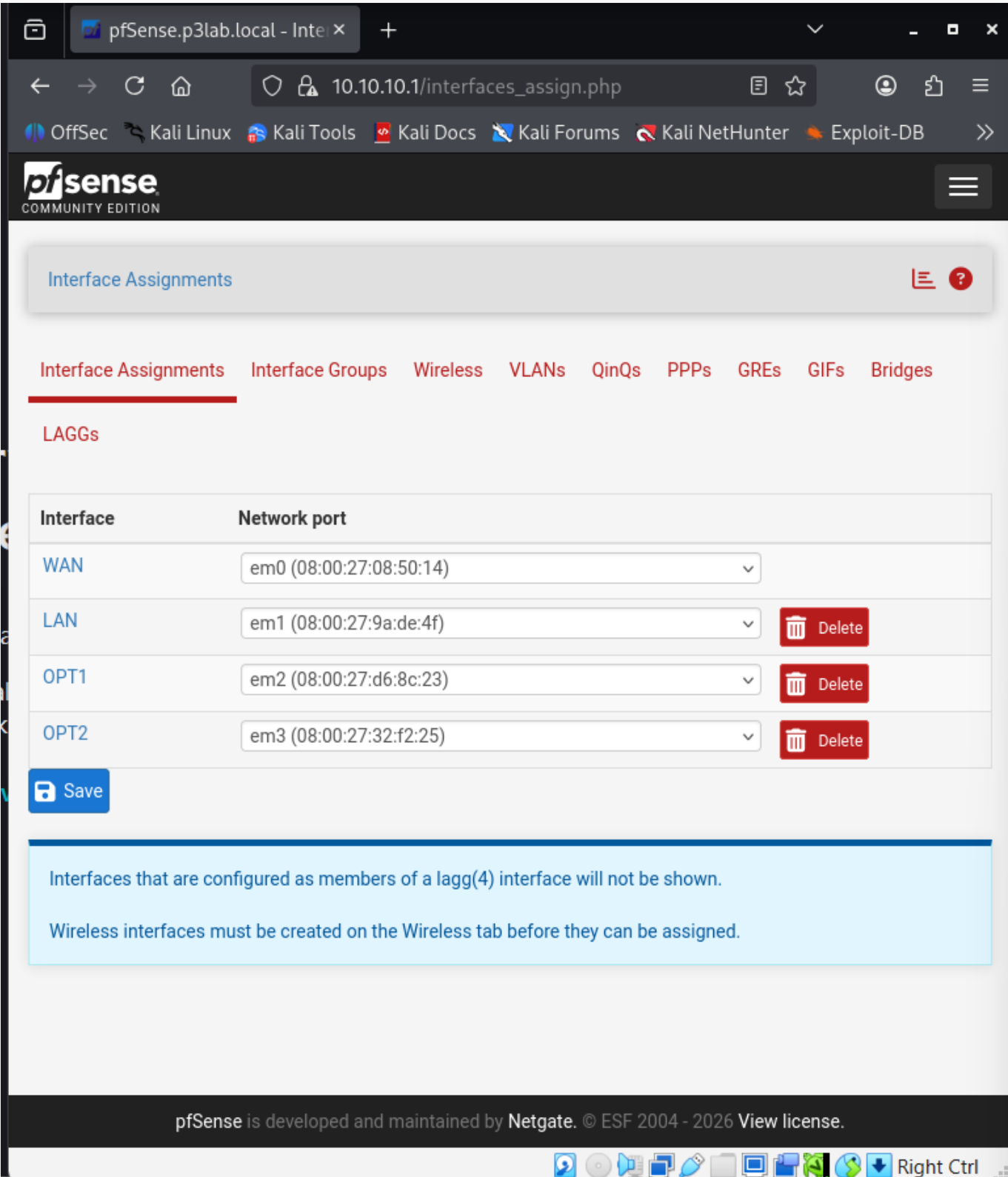
Confirmed interface assignments, static gateway IPs, and DHCP scopes for segmented clients.

Evidence screenshots in this section: 11

- 3A-pfsense-interface-assignments-em0-em1-em2-em3
- 3B-1-pfsense-db-interface-general-settings
- 3B-2-pfsense-db-interface-static-ip-settings
- 3C-1-pfsense-admin-interface-general-settings
- 3C-2-pfsense-admin-interface-static-ip-settings
- 3D-1-pfsense-web-lan-dhcp-general-settings
- 3D-2-pfsense-web-lan-dhcp-address-pool
- 3E-1-pfsense-db-dhcp-general-settings
- 3E-2-pfsense-db-dhcp-address-pool
- 3F-1-pfsense-admin-dhcp-general-settings (2)
- 3F-2-pfsense-admin-dhcp-address-pool

Interface Assignments and DHCP

3A pfSense interface assignments em0 em1 em2 em3



Interface Assignments and DHCP

3B 1 pfsense db interface general settings

 COMMUNITY EDITION

DB (em2)

The changes have been applied successfully.

General Configuration

Enable

☒ Enable interface

Description

DB

Enter a description (name) for the interface here.

IPv4 Configuration Type

Static IPv4

IPv6 Configuration Type

None

MAC Address

XX:XX:XX:XX:XX:XX

This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some

Interface Assignments and DHCP

3B 2 pfsense db interface static ip settings

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex

Explicitly set speed and duplex mode for this interface.

WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address

 /

IPv4 Upstream gateway

[+ Add a new gateway](#)

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.

On local area network interfaces the upstream gateway should be "none".

Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).

Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses

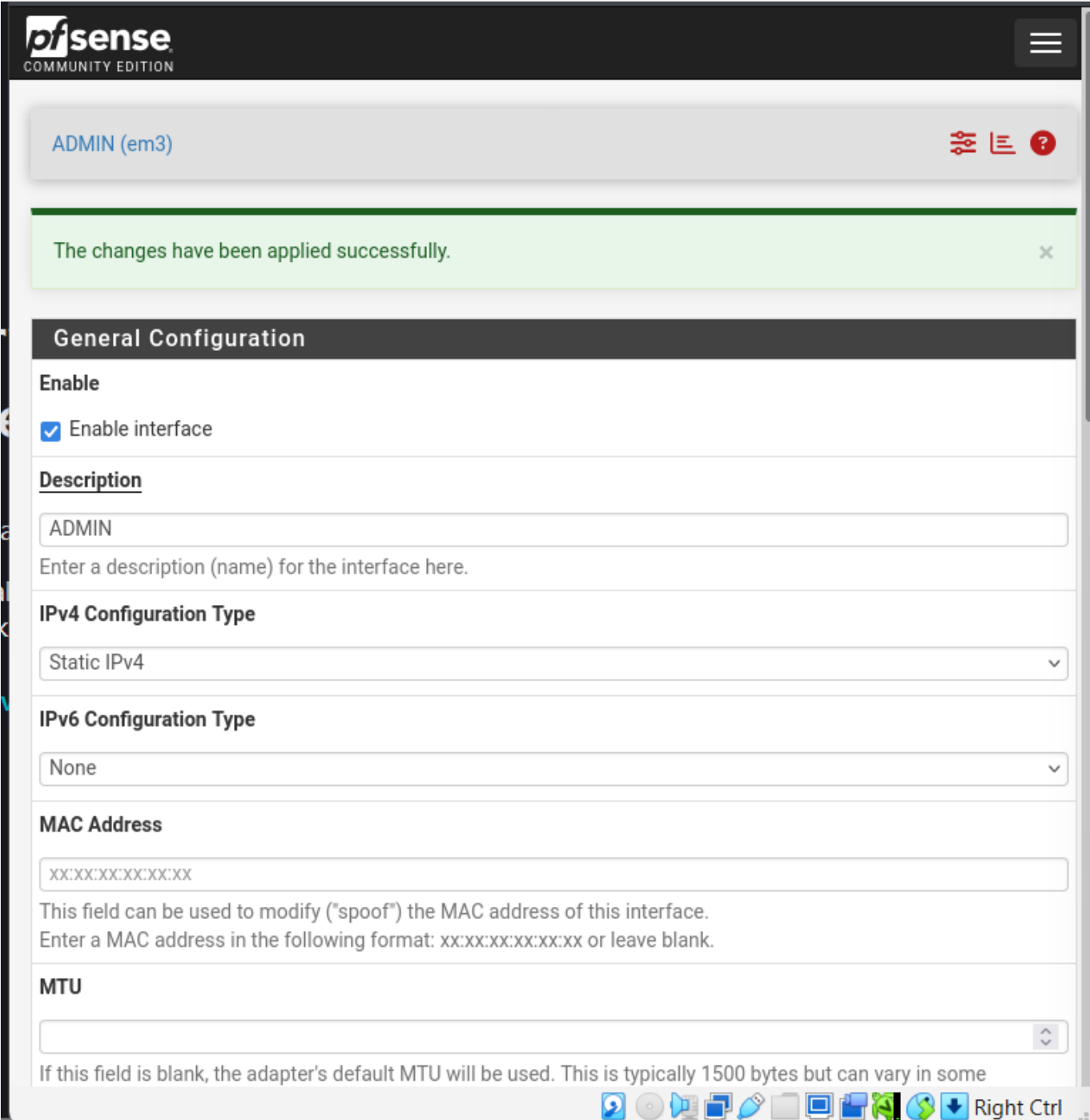
☐

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks

Interface Assignments and DHCP

3C 1 pfsense admin interface general settings



Interface Assignments and DHCP

3C 2 pfsense admin interface static ip settings

circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex

Default (no preference, typically autoselect)

▼

Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address

10.10.30.1

/

24

▼

IPv4 Upstream gateway

None

▼

+

Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses

☐

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Interface Assignments and DHCP

3D 1 pfsense web lan dhcp general settings

Settings LAN DB ADMIN

General Settings

DHCP Backend

Kea DHCP

Enable

☒ Enable DHCP server on LAN interface

Deny Unknown Clients

Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

Ignore Client Identifiers

☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request

This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

DNS Registration

Track server

Optionally overrides the DHCP server default DNS registration policy to force a specific policy.

Early DNS Registration

Track server

Optionally overrides the DHCP server default early DNS registration policy to force a specific policy.

Primary Address Pool

Subnet

Interface Assignments and DHCP

3D 2 pfsense web lan dhcp address pool

← → ↻ 🏠 🔒 10.10.10.1/services_dhcp.php?if=lan ☆ 👤 📁 ☰

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB >>

address. Note that the resulting server behavior violates the official DHCP specification.

DNS Registration

Track server ▾

Optionally overrides the DHCP server default DNS registration policy to force a specific policy.

Early DNS Registration

Track server ▾

Optionally overrides the DHCP server default early DNS registration policy to force a specific policy.

Primary Address Pool

Subnet

10.10.10.0/24

Subnet Range

10.10.10.1 - 10.10.10.254

Address Pool Range

10.10.10.100

From

10.10.10.199

To

The specified range for this pool must not be within the range configured on any other address pool for this interface.

Additional Pools

[+ Add Address Pool](#)

If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Server Options

Interface Assignments and DHCP

3E 1 pfsense db dhcp general settings

DB



The changes have been applied successfully.

Settings

LAN

DB

ADMIN

General Settings

DHCP Backend

Kea DHCP

Enable

☒ Enable DHCP server on DB interface

Deny Unknown Clients

Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

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DNS Registration

Interface Assignments and DHCP

3E 2 pfsense db dhcp address pool

address. Note that the resulting server behavior violates the official DHCP specification.

DNS Registration

Track server

Optionally overrides the DHCP server default DNS registration policy to force a specific policy.

Early DNS Registration

Track server

Optionally overrides the DHCP server default early DNS registration policy to force a specific policy.

Primary Address Pool

Subnet

10.10.20.0/24

Subnet Range

10.10.20.1 - 10.10.20.254

Address Pool Range

10.10.20.100

From

10.10.20.199

To

The specified range for this pool must not be within the range configured on any other address pool for this interface.

Additional Pools

+ Add Address Pool

If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Server Options

3E-2-pfsense-db-dhcp-address-pool.png

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Interface Assignments and DHCP

3F 1 pfsense admin dhcp general settings

 pfSense
COMMUNITY EDITION



ADMIN



The changes have been applied successfully.

Settings LAN DB ADMIN

General Settings

DHCP Backend

Kea DHCP

Enable

☒ Enable DHCP server on ADMIN interface

Deny Unknown Clients

Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

Ignore Client Identifiers

☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request

This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

DNS Registration

Interface Assignments and DHCP

3F 2 pfsense admin dhcp address pool

Primary Address Pool

Subnet

10.10.30.0/24

Subnet Range

10.10.30.1 - 10.10.30.254

Address Pool Range

10.10.30.100

From

10.10.30.199

To

The specified range for this pool must not be within the range configured on any other address pool for this interface.

Additional Pools

+ Add Address Pool

If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Server Options

WINS Servers

WINS Server 1

WINS Server 2

DNS Servers

10.10.30.1

Test 04-Firewall-Rules-and-Segmentation-Policy

Firewall Rules and Segmentation Policy

Control Objective

Implemented aliases and firewall rules to separate WEB, DB, and ADMIN network traffic.

Evidence screenshots in this section: 7

- 4A-pfsense-firewall-network-aliases
- 4B-1-pfsense-firewall-reload-status-lan-rules
- 4B-2-pfsense-lan-web-segmentation-rules
- 4C-pfsense-db-isolation-rules
- 4D-pfsense-admin-management-access-rule
- 4E-1-pfsense-filter-reload-status-start
- 4E-2-pfsense-filter-reload-status-complete

Firewall Rules and Segmentation Policy

4A pfsense firewall network aliases

 pfSense
COMMUNITY EDITION



IP



The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.



IP

Ports

URLs

All

Firewall Aliases IP

Name	Type	Values	Description	Actions
ADMIN_NET	Network(s)	10.10.30.0/24	ADMIN subnet for Project 3 segmentation	  
DB_NET	Network(s)	10.10.20.0/24	DB subnet for Project 3 segmentation	  
WEB_NET	Network(s)	10.10.10.0/24	WEB/LAN subnet for Project 3 segmentation	  

 Add

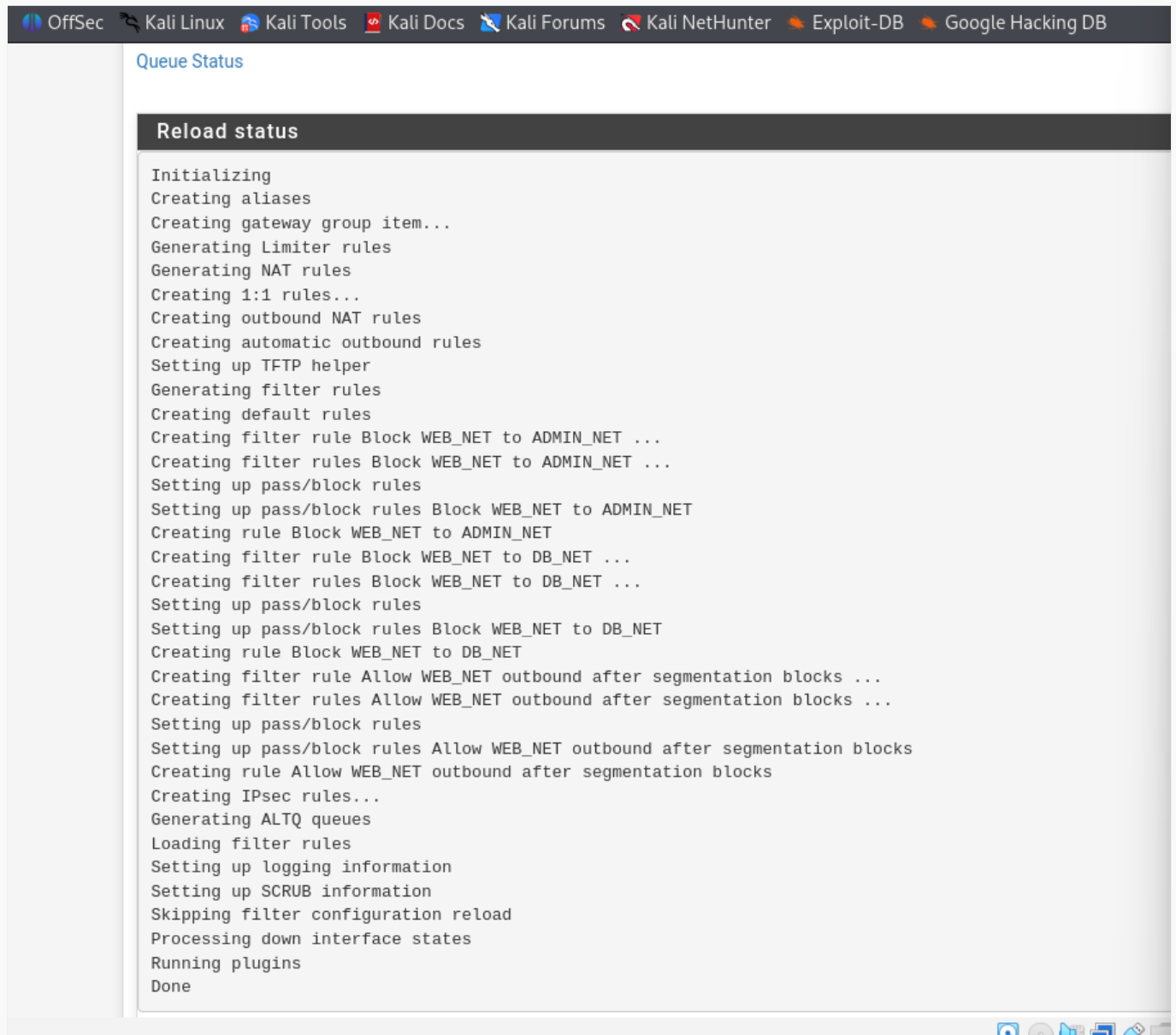
 Import



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Firewall Rules and Segmentation Policy

4B 1 pfsense firewall reload status lan rules



Firewall Rules and Segmentation Policy

4B 2 pfsense lan web segmentation rules

 **System** ▾ **Interfaces** ▾ **Firewall** ▾ **Services** ▾ **VPN** ▾ **Status** ▾ **Diagnostics** ▾ **Help** ▾

Firewall / Rules / LAN

Floating

WAN

LAN

DB

ADMIN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Action
☒	✓ 0/1.21 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	✗ 0/0 B	IPv4 *	WEB_NET	*	ADMIN_NET	*	*	none		Block WEB_NET to ADMIN_NET	 
☐	✗ 0/0 B	IPv4 *	WEB_NET	*	DB_NET	*	*	none		Block WEB_NET to DB_NET	 
☐	✓ 2/536 B	IPv4 *	WEB_NET	*	*	*	*	none		Allow WEB_NET outbound after segmentation blocks	 
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	  

 Add

 Add

 Delete

 Toggle

 Copy

 Save





Firewall Rules and Segmentation Policy

4C pfsense db isolation rules

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Firewall / Rules / DB

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

Floating WAN LAN DB ADMIN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0/0 B	IPv4 *	DB_NET	*	ADMIN_NET	*	*	none		Block DB_NET to ADMIN_NET	  
☐	✗ 0/0 B	IPv4 *	DB_NET	*	WEB_NET	*	*	none		Block DB_NET to WEB_NET	  

↑ Add

↓ Add

 Delete

 Toggle

 Copy

 Save

+



Firewall Rules and Segmentation Policy

4D pfsense admin management access rule

Firewall / Rules / ADMIN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

Floating WAN LAN DB ADMIN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 *	ADMIN_	*	*	*	*	none	Allow ADMIN_NET controlled management access	   

↑ Add

↓ Add

🗑 Delete

🔄 Toggle

📄 Copy

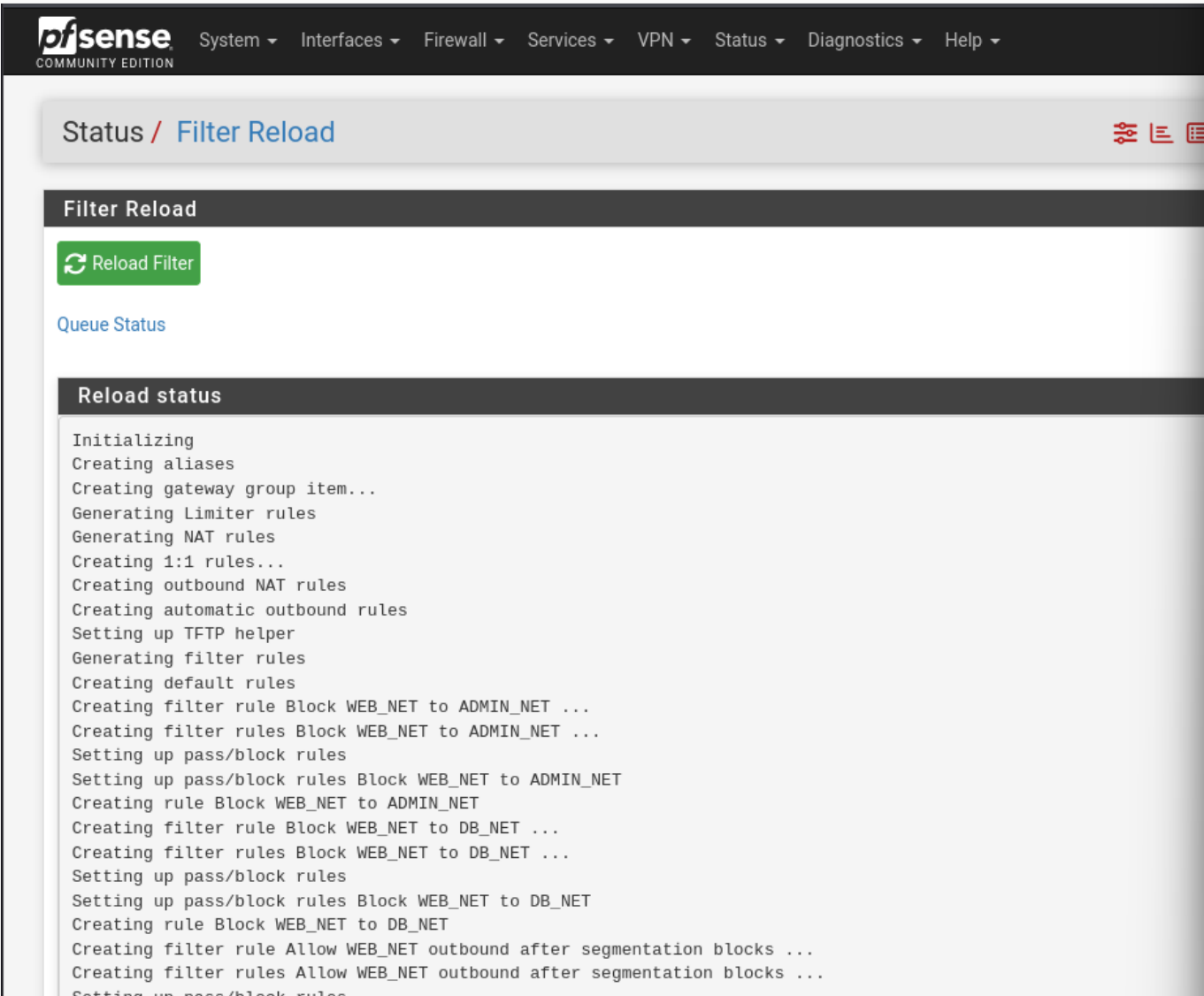
💾 Save

+ Se



Firewall Rules and Segmentation Policy

4E 1 pfsense filter reload status start



4E-1-pfsense-filter-reload-status-start.png

Firewall Rules and Segmentation Policy

4E 2 pfsense filter reload status complete

```
Creating filter rule Block WEB_NET to ADMIN_NET ...
Creating filter rules Block WEB_NET to ADMIN_NET ...
Setting up pass/block rules
Setting up pass/block rules Block WEB_NET to ADMIN_NET
Creating rule Block WEB_NET to ADMIN_NET
Creating filter rule Block WEB_NET to DB_NET ...
Creating filter rules Block WEB_NET to DB_NET ...
Setting up pass/block rules
Setting up pass/block rules Block WEB_NET to DB_NET
Creating rule Block WEB_NET to DB_NET
Creating filter rule Allow WEB_NET outbound after segmentation blocks ...
Creating filter rules Allow WEB_NET outbound after segmentation blocks ...
Setting up pass/block rules
Setting up pass/block rules Allow WEB_NET outbound after segmentation blocks
Creating rule Allow WEB_NET outbound after segmentation blocks
Creating filter rule Block DB_NET to ADMIN_NET ...
Creating filter rules Block DB_NET to ADMIN_NET ...
Setting up pass/block rules
Setting up pass/block rules Block DB_NET to ADMIN_NET
Creating rule Block DB_NET to ADMIN_NET
Creating filter rule Block DB_NET to WEB_NET ...
Creating filter rules Block DB_NET to WEB_NET ...
Setting up pass/block rules
Setting up pass/block rules Block DB_NET to WEB_NET
Creating rule Block DB_NET to WEB_NET
Creating filter rule Allow ADMIN_NET controlled management access ...
Creating filter rules Allow ADMIN_NET controlled management access ...
Setting up pass/block rules
Setting up pass/block rules Allow ADMIN_NET controlled management access
Creating rule Allow ADMIN_NET controlled management access
Creating IPsec rules...
Generating ALTQ queues
Loading filter rules
Setting up logging information
Setting up SCRUB information
Skipping filter configuration reload
Processing down interface states
Running plugins
Done
```

Test 05-Allowed-Traffic-Validation

Allowed Traffic Validation

Control Objective

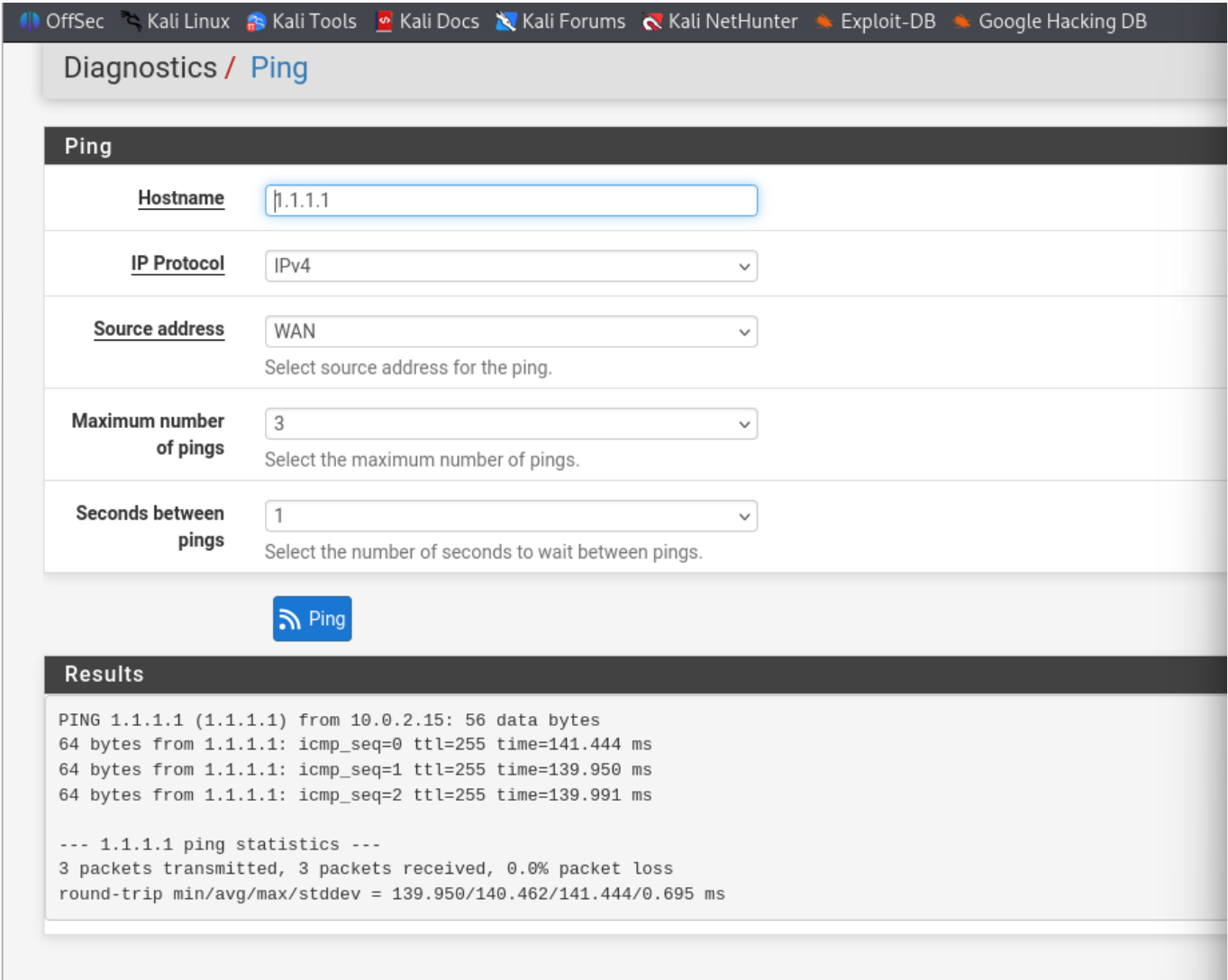
Validated required outbound connectivity, WAN pings, and DNS resolution after troubleshooting.

Evidence screenshots in this section: 4

- 5A-1-pfsense-wan-ping-1.1.1.1-success
- 5A-2-pfsense-wan-ping-8.8.8.8-success
- 5B-pfsense-dns-lookup-google-success
- 5C-kali-web-dns-resolution-success

Allowed Traffic Validation

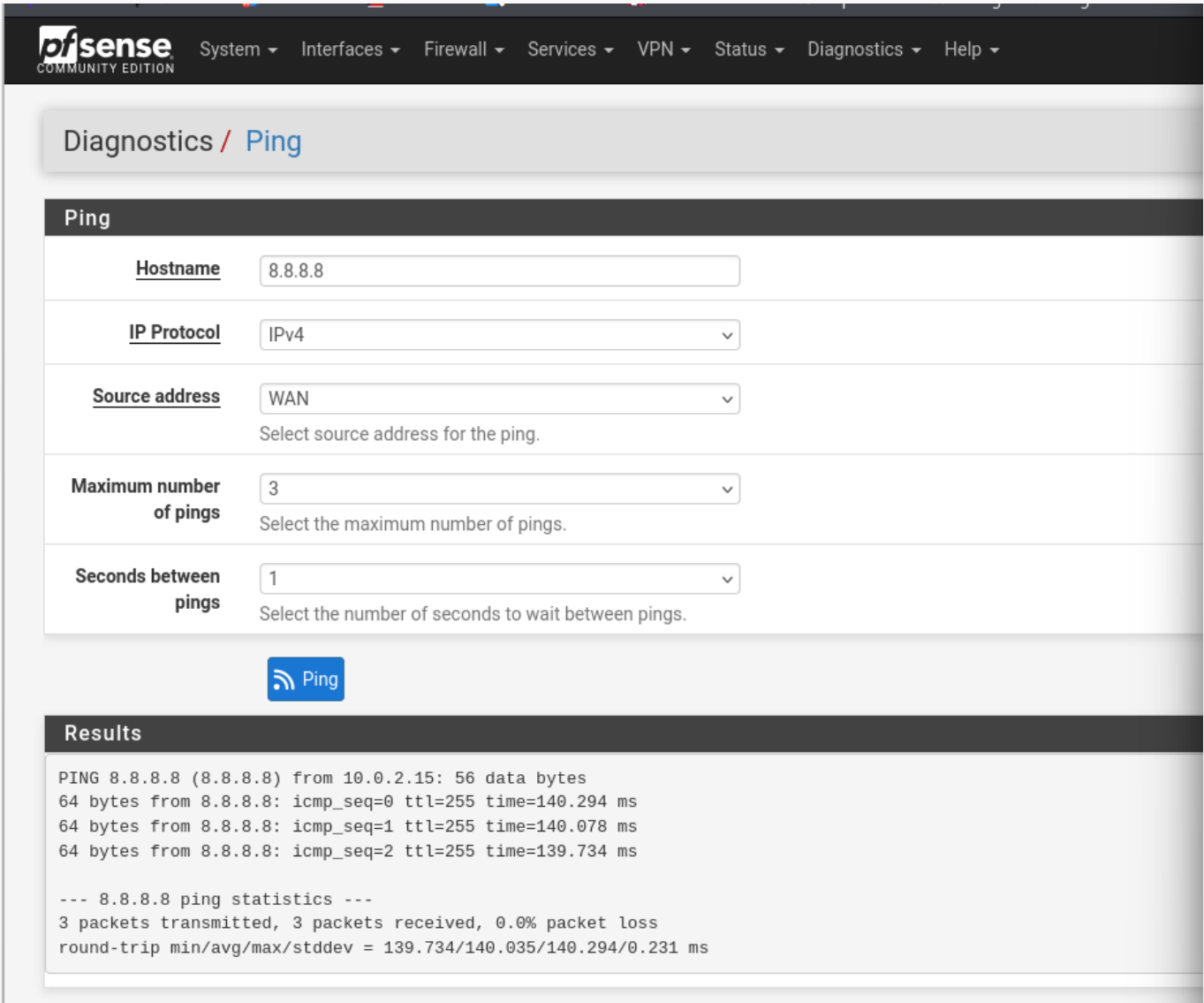
5A 1 pfsense wan ping 1.1.1.1 success



5A-1-pfsense-wan-ping-1.1.1.1-success.png

Allowed Traffic Validation

5A 2 pfsense wan ping 8.8.8.8 success



Allowed Traffic Validation

5B pfSense dns lookup google success

SystemInterfacesFirewallServicesVPNStatusDiagnosticsHelp

COMMUNITY EDITION

Diagnostics / DNS Lookup

DNS Lookup

Hostname

google.com

Lookup

Add Alias

Results

Result	Record type
216.58.223.206	A
2c0f:fb50:4003:802::200e	AAAA

Timings

Name server	Query time
127.0.0.1	1717 msec
::1	4 msec
1.1.1.1	No response
8.8.8.8	No response
10.0.2.3	127 msec

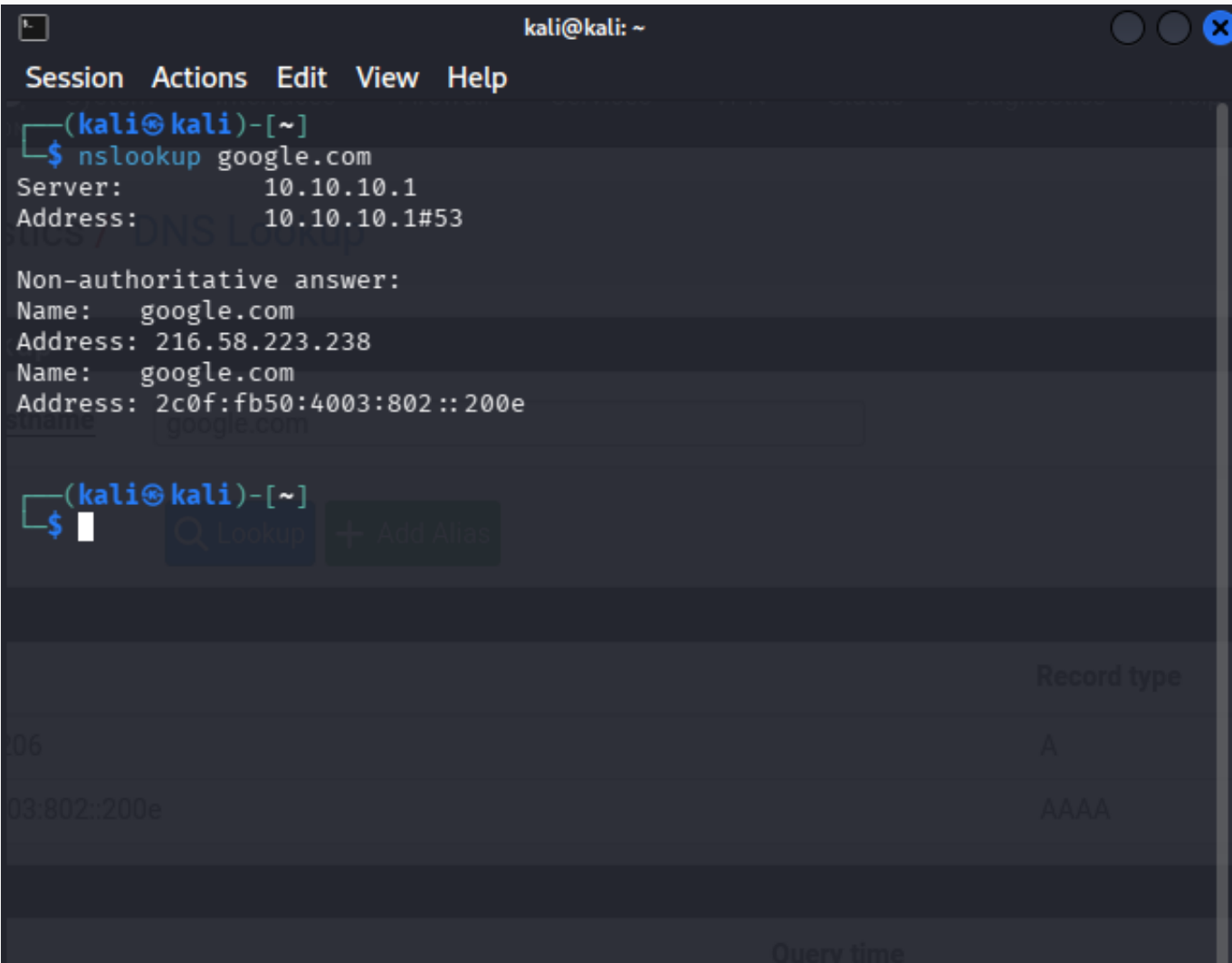
More Information

Ping

Traceroute

Allowed Traffic Validation

5C kali web dns resolution success



Test 06-Blocked-Lateral-Movement-Validation

Blocked Lateral Movement Validation

Control Objective

Proved WEB client traffic to DB and ADMIN gateways was blocked and logged.

Evidence screenshots in this section: 3

- 6A-kali-web-blocked-ping-to-db-and-admin
- 6B-pfsense-lan-rules-block-web-to-db-admin
- 6C-pfsense-firewall-logs-blocked-web-to-db-admin

Blocked Lateral Movement Validation

6A kali web blocked ping to db and admin

```
(kali@kali)-[~]
$ ping -c 4 10.10.20.1 dd Alias
ping -c 4 10.10.30.1
PING 10.10.20.1 (10.10.20.1) 56(84) bytes of data.

— 10.10.20.1 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3065ms
Record type

PING 10.10.30.1 (10.10.30.1) 56(84) bytes of data.
A

— 10.10.30.1 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3893ms
AAAA

(kali@kali)-[~]
```

6B pfsense lan rules block web to db admin



Blocked Lateral Movement Validation

6C pfsense firewall logs blocked web to db admin

Normal ViewDynamic ViewSummary View

Last 18 Firewall Log Entries. (Maximum 500)							
Action	Time	Interface	Rule	Source	Destination	Protocol	
✗	Sep 11 17:23:44	LAN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 14:46:33	LAN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 14:46:33	LAN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 17:45:19	LAN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 17:45:20	LAN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 16:04:51	DB	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 16:04:51	DB	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 16:04:52	DB	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 16:30:57	ADMIN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 12 16:30:57	ADMIN	(@4294967295)	i-::	i+[ff02::16]	Opti	
✗	Sep 13 13:23:15	LAN	Block WEB_NET to DB_NET (1789238716)	i-10.10.10.100	i+10.10.20.1	ICM	
✗	Sep 13 13:23:16	LAN	Block WEB_NET to DB_NET (1789238716)	i-10.10.10.100	i+10.10.20.1	ICM	
✗	Sep 13 13:23:17	LAN	Block WEB_NET to DB_NET (1789238716)	i-10.10.10.100	i+10.10.20.1	ICM	
✗	Sep 13 13:23:18	LAN	Block WEB_NET to DB_NET (1789238716)	i-10.10.10.100	i+10.10.20.1	ICM	
✗	Sep 13 13:23:29	LAN	Block WEB_NET to ADMIN_NET (1789239004)	i-10.10.10.100	i+10.10.30.1	ICM	
✗	Sep 13 13:23:30	LAN	Block WEB_NET to ADMIN_NET (1789239004)	i-10.10.10.100	i+10.10.30.1	ICM	
✗	Sep 13 13:23:31	LAN	Block WEB_NET to ADMIN_NET (1789239004)	i-10.10.10.100	i+10.10.30.1	ICM	
✗	Sep 13 13:23:32	LAN	Block WEB_NET to ADMIN_NET (1789239004)	i-10.10.10.100	i+10.10.30.1	ICM	

Test 07-Admin-Access-Hardening

Admin Access Hardening

Control Objective

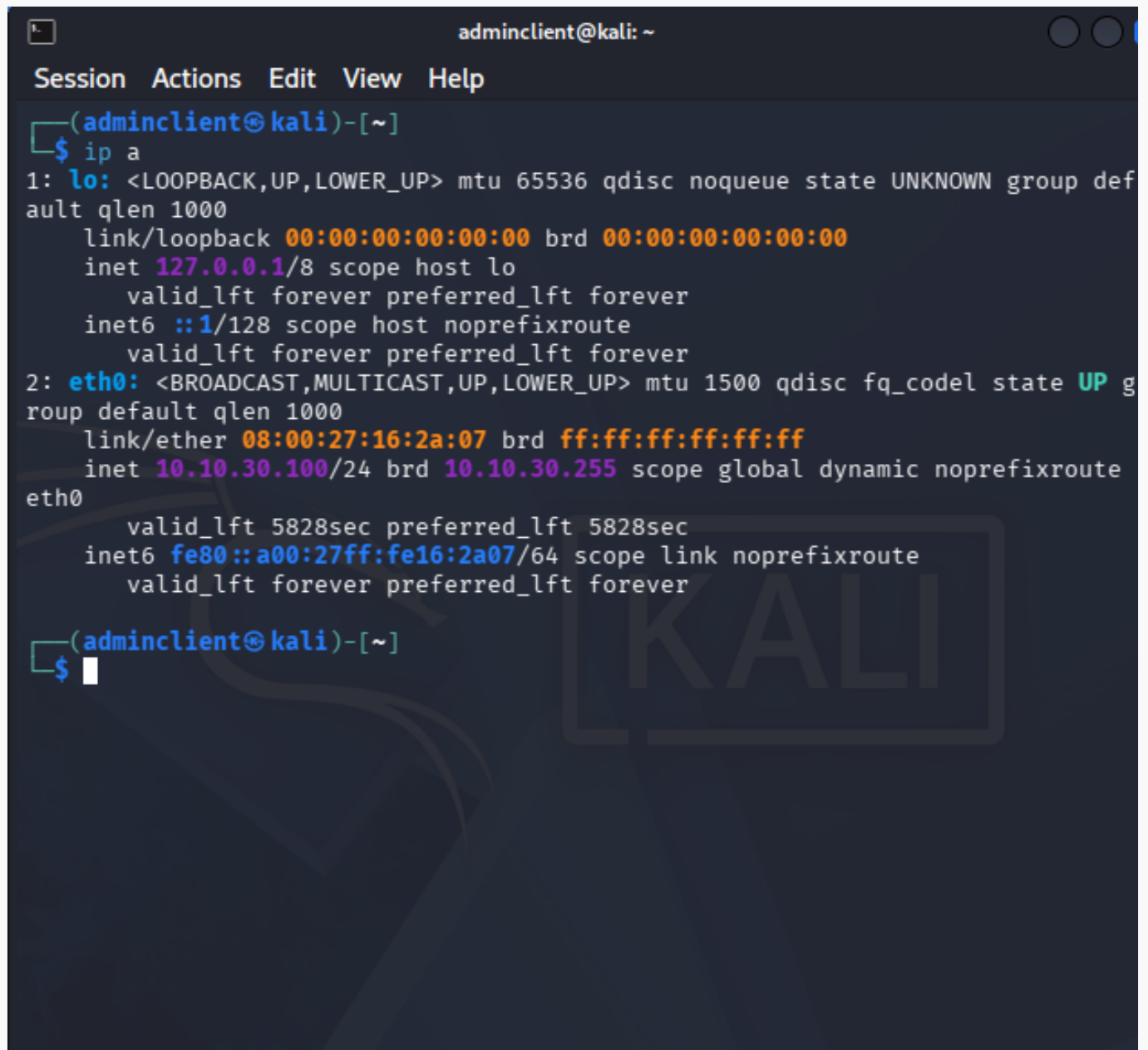
Restricted pfSense management to the ADMIN network and blocked WEB access to firewall management.

Evidence screenshots in this section: 5

- 7A admin client DHCP address validation
- 7B-admin-client-pfsense-admin-access-success
- 7C-pfsense-admin-access-hardening-settings-applied
- 7D-pfsense-lan-rule-block-web-management
- 7E-kali-web-pfsense-management-blocked

Admin Access Hardening

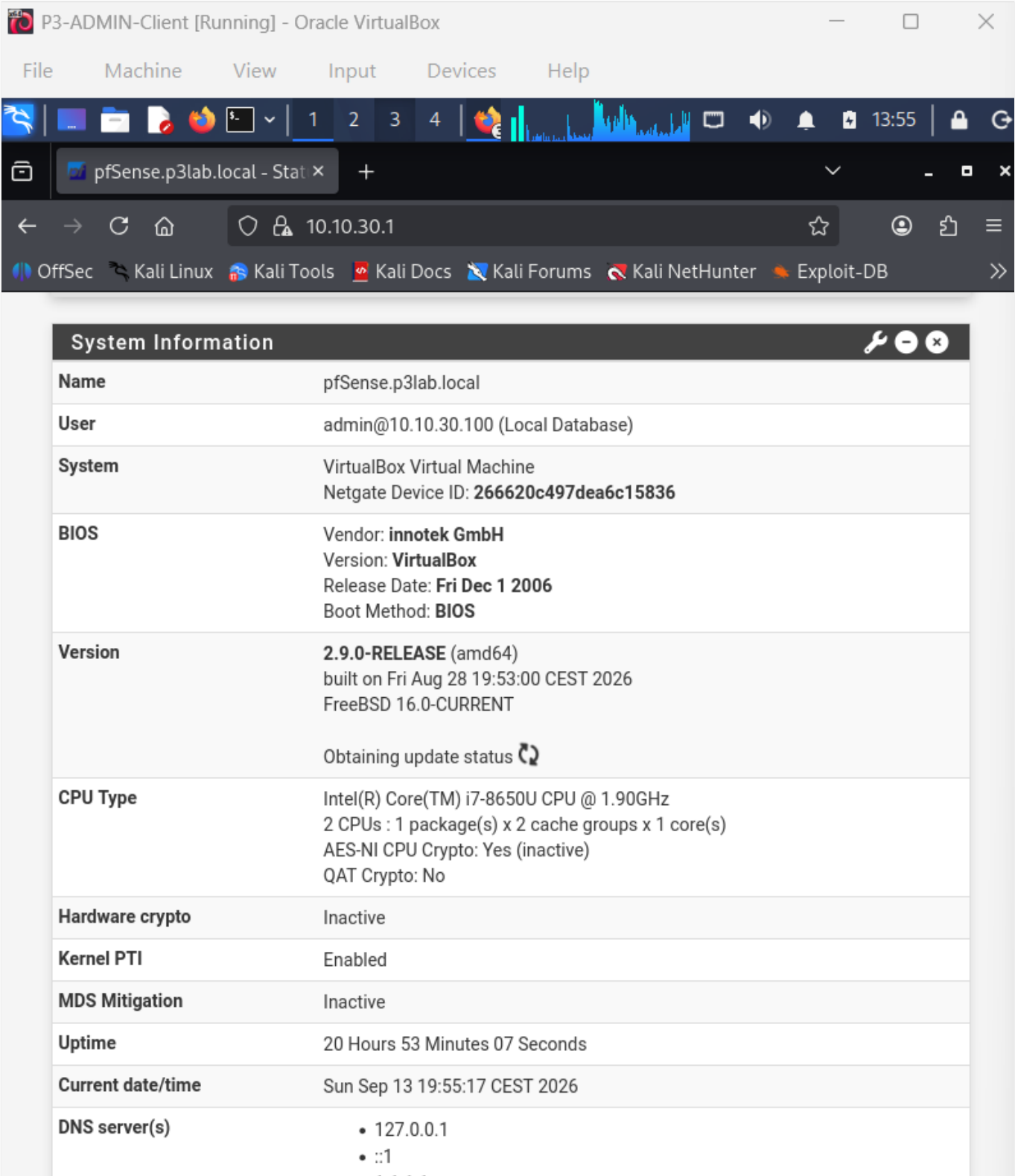
7A admin client DHCP address validation



```
adminclient@kali: ~  
Session Actions Edit View Help  
(adminclient@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:16:2a:07 brd ff:ff:ff:ff:ff:ff  
    inet 10.10.30.100/24 brd 10.10.30.255 scope global dynamic noprefixroute  
eth0  
    valid_lft 5828sec preferred_lft 5828sec  
    inet6 fe80::a00:27ff:fe16:2a07/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
(adminclient@kali)-[~]  
$
```

Admin Access Hardening

7B admin client pfsense admin access success



7B-admin-client-pfsense-admin-access-success.png

Admin Access Hardening

7C pfSense admin access hardening settings applied

COMMUNITY EDITION



System / Advanced / Admin Access



The changes have been applied successfully.
One moment...redirecting to https://10.10.30.1/system_advanced_admin.php in 20 seconds.



Admin Access

Firewall & NAT

Networking

Miscellaneous

System Tunables

Notifications

webConfigurator

Protocol

☐ HTTP

☒ HTTPS (SSL/TLS)

No Certificates have been defined. A certificate is required before SSL/TLS can be enabled.
[Create or Import](#) a Certificate.

SSL/TLS Certificate

GUI default (6aa41d2731faf)



Certificates known to be incompatible with use for HTTPS are not included in this list, such as certificates using incompatible ECDSA curves or weak digest algorithms.

TCP port



Enter a custom port number for the webConfigurator above to override the default (80 for HTTP, 443 for HTTPS). Changes will take effect immediately after save.

Max Processes

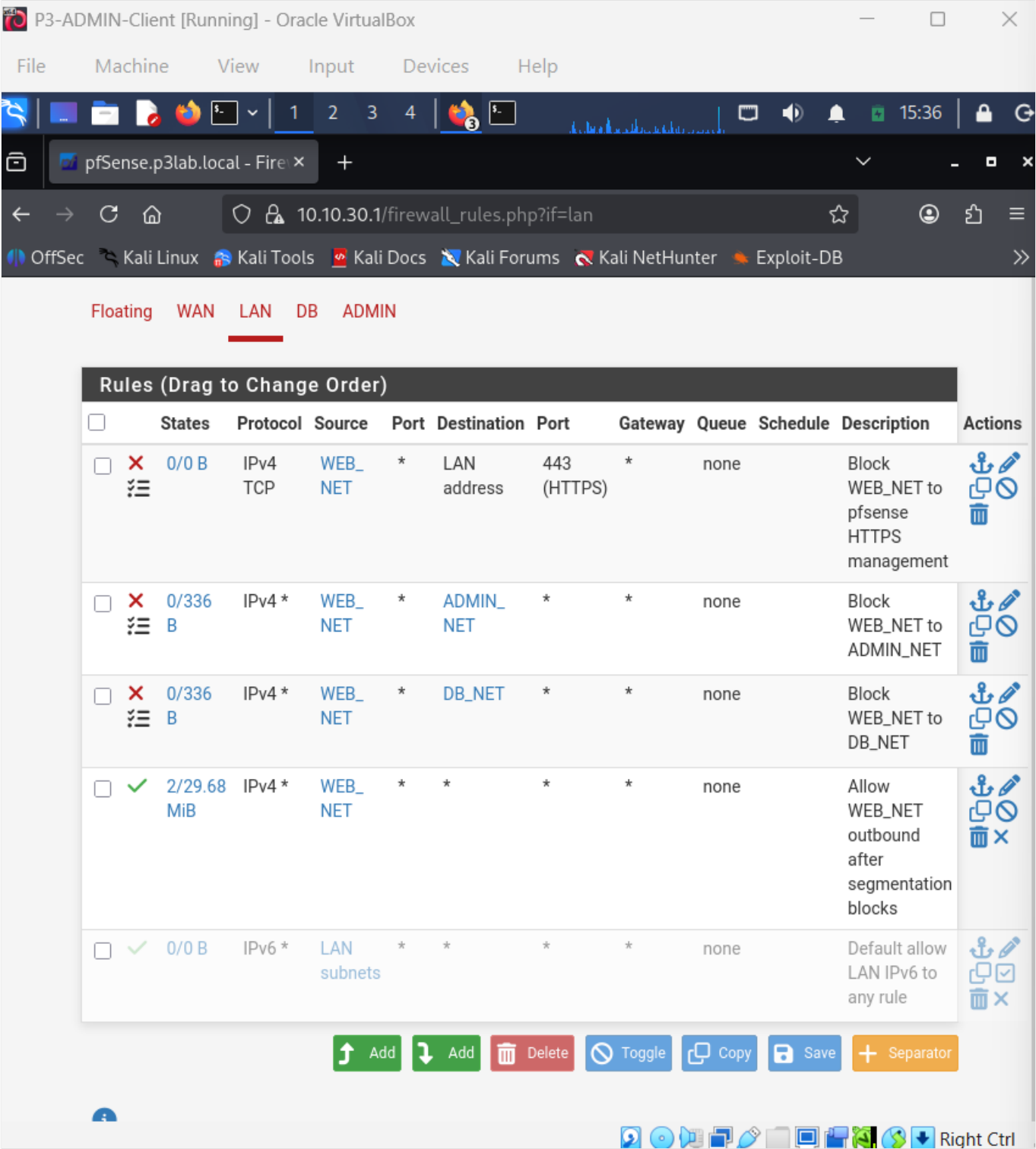
2



Enter the number of webConfigurator processes to run. This defaults to 2. Increasing this will allow more users/browsers to access the GUI concurrently.

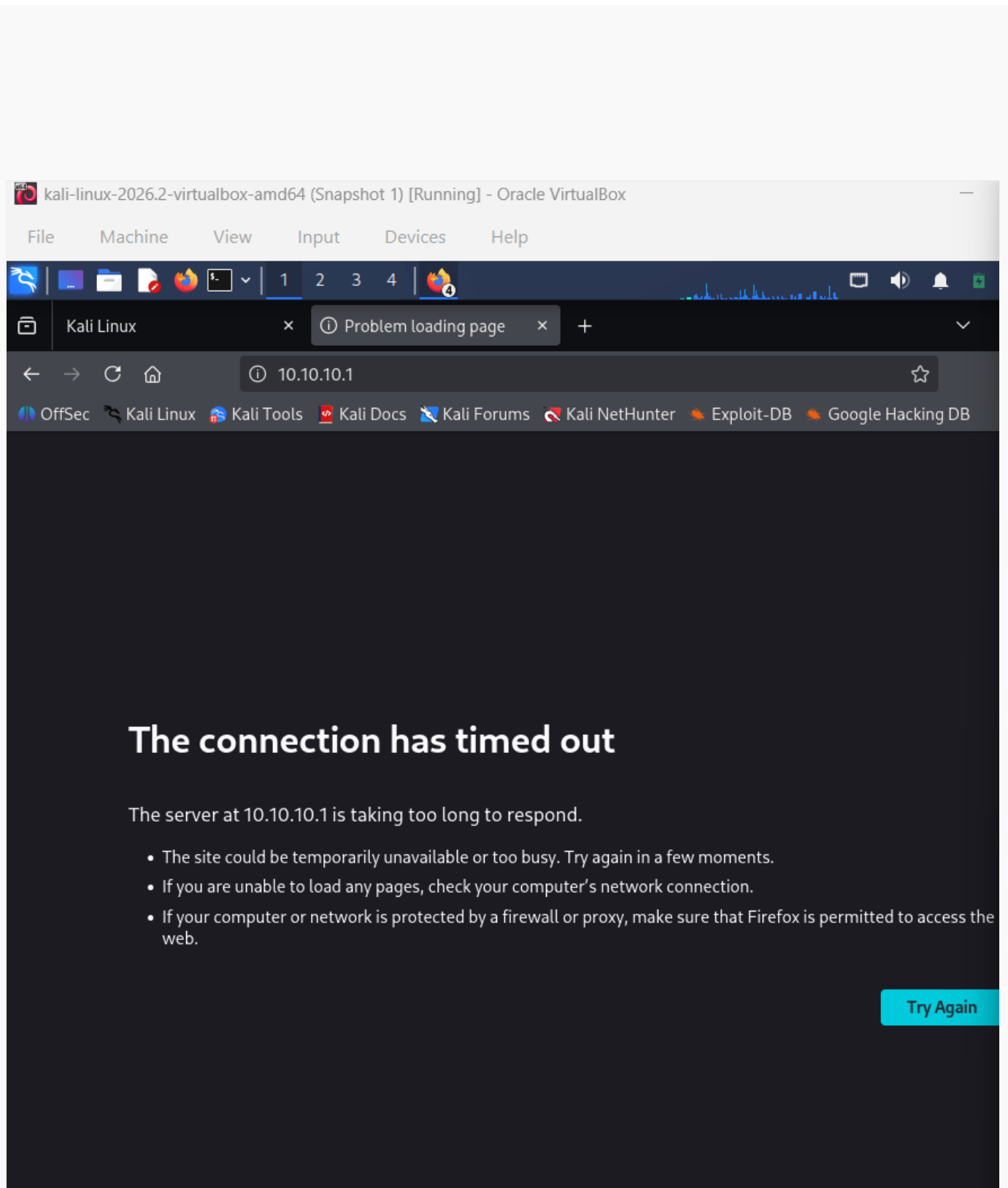
Admin Access Hardening

7D pfSense lan rule block web management



Admin Access Hardening

7E kali web pfsense management blocked



Test 08-Logging-and-Monitoring-Evidence

Logging and Monitoring Evidence

Control Objective

Captured firewall log evidence and monitoring visibility for blocked management attempts.

Evidence screenshots in this section: 4

- 8A-1-pfsense-firewall-log-web-management-blocked
- 8A-2-pfsense-firewall-log-web-management-blocked
- 8B-1-pfsense-monitoring-dashboard
- 8B-2-pfsense-monitoring-dashboard

Logging and Monitoring Evidence

8A 1 pfsense firewall log web management blocked

SystemFirewallDHCPAuthenticationIPsecPPPPPPoE/L2TP ServerOpenVPNNTPPackagesSettings

Normal ViewDynamic ViewSummary View

Last 74 Firewall Log Entries. (Maximum 500)						
Action	Time	Interface	Rule	Source	Destination	Protocol
✗	Sep 11 17:23:44	LAN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 14:46:33	LAN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 14:46:33	LAN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 17:45:19	LAN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 17:45:20	LAN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 16:04:51	DB	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 16:04:51	DB	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 16:04:52	DB	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 16:30:57	ADMIN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 12 16:30:57	ADMIN	(@4294967295)	i [::]	i+ [ff02::16]	OpenSSH
✗	Sep 13 13:23:15	LAN	Block WEB_NET to DB_NET (1789238716)	i 10.10.10.100	i+ 10.10.20.1	ICMP

Logging and Monitoring Evidence

8A 2 pfsense firewall log web management blocked

Sep 13 21:39:44	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42052	  10.10.10.1:443	TCP:S
Sep 13 21:39:44	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42066	  10.10.10.1:443	TCP:S
Sep 13 21:40:00	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42052	  10.10.10.1:443	TCP:S
Sep 13 21:40:00	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42066	  10.10.10.1:443	TCP:S
Sep 13 21:40:34	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42066	  10.10.10.1:443	TCP:S
Sep 13 21:40:34	LAN	 Block WEB_NET to pfsense HTTPS management (1789325576)	  10.10.10.100:42052	  10.10.10.1:443	TCP:S

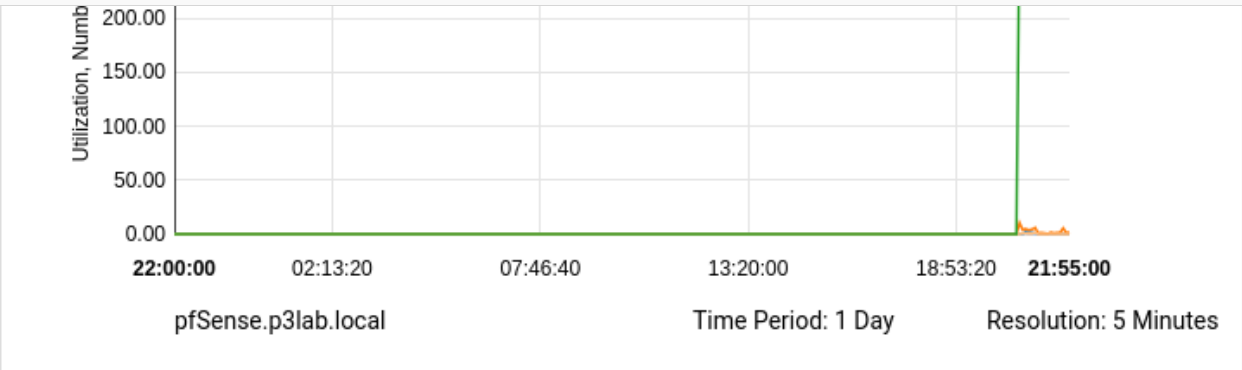
Logging and Monitoring Evidence

8B 1 pfsense monitoring dashboard



Logging and Monitoring Evidence

8B 2 pfsense monitoring dashboard



Data Summary					
	Minimum	Average	Maximum	Last	95th Percentile
user util.	0.00 %	0.33 %	2.04 %	0.00 %	
nice util.	0.00 %	0.07 %	0.28 %	0.00 %	
system util.	0.28 %	2.92 %	10.51 %	1.58 %	
interrupt	0.00 %	0.29 %	1.50 %	0.09 %	
processes	297.00	297.01	297.22	297.00	



Test 09-Final-Risk-Reduction-Summary

Final Risk Reduction Summary

Control Objective

Summarized the final security controls and risk reduction achieved by the lab.

Evidence screenshots in this section: 1

- 9A-project3-final-risk-reduction-summary

Final Risk Reduction Summary

9A project3 final risk reduction summary

